

XRPL Validator — Daily Engineering Report • 2026-08-01

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 6 commits (xrpl-validator 6)

Engineering changes

- **fix(ledger): consolidate NFToken pages when a removal shrinks one** (761df02 , xrpl-validator) `page_remove` handled an EMPTIED page (delete, relink neighbours) but did nothing when a page merely shrank. `rippled` consolidates on every removal (NFTokenHelpers.cpp `removeToken`): "The current page isn't empty. Update it and then try to consolidate pages. Note that this consolidation attempt may actually merge three pages into one!"
- **fix(ledger): split a full NFToken page before inserting** (9a82325 , xrpl-validator) `page_insert` had NO capacity check: it located the page and inserted, so a full page silently grew to 33 tokens — past `kDirMaxTokensPerPage`. `rippled` splits first (NFTokenHelpers.cpp `getPageForToken`): the lower part moves to a NEW page keyed below, the rest stays, and the predecessor is relinked. That Created + Modified pair is exactly what mainnet emits.
- **fix(probe): hydrate trust lines for the currencies named in Paths** (549abb1 , xrpl-validator) We hydrated only the Amount and SendMax currencies, never the intermediate hops. A payment that ripples through a third currency therefore saw NO trust line for that hop, ran `trustCreate` for a line that already exists, and invented a directory page to hold it.
- **fix(ledger): align IOU addition the way STAmount does** (d5a08c8 , xrpl-validator) `STAmount operator+` brings both operands to the LARGER exponent by integer division: `while (ov1 < ov2) { vv1 /= 10; ++ov1; }` `while (ov2 < ov1) { vv2 /= 10; ++ov2; }`
- **fix(ledger): apply tfSetfAuth to the trust line** (4bac28e , xrpl-validator) We honoured `tfSetfAuth` in preclaim's redundancy test but never APPLIED it. `do_apply` handled `tfSetNoRipple` / `tfClearNoRipple` / `tfSetFreeze` / `tfClearFreeze` and had no branch for the auth bit, so a `TrustSet` whose only effect is authorisation left the line untouched entirely.
- **fix(ledger): CredentialCreate joins both owner directories** (be244f0 , xrpl-validator) A credential is inserted into the issuer's AND the subject's owner directory, but is owned only by the issuer — `CredentialCreate.cpp`:147-174: "Added to both dirs, owned only by issuer. `CredentialAccept` will transfer ownership to subject." So the issuer's `dirInsert` carries the single `adjustOwnerCount(+1)` and the subject's adds no reserve at all. A self-issued credential (`subject == account`) skips the subject insert ...

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Credential-Gated Auto-Preauth for Tokens** (compliance) Add an optional sfAllowedCredentials SField (STArray, max 8 entries of {Issuer, CredentialType}) to MPTokenIssuance, usable only when lsfMPTRequireAut
- **Native NFT Loan Delegation Without Remint** (nft) Add two SFields to NFTokenPage entries, sfBorrower (AccountID) and sfLoanExpiration (LedgerIndex), settable only via a new NFTokenLend transaction (ow
- **Deterministic Scheduled & Oracle-Triggered Execution** (hooks) Add a new owner-reserved ledger object ltSCHEDULED_TX (keylet::scheduledTx(account, sequence)) created by ScheduledTxCreate, holding a pre-signed inne

Appendix — commits

Time	Repo	Commit	Subject
01:56	xrpl-validator	761df02	fix(ledger): consolidate NFToken pages when a removal shrinks one
14:07	xrpl-validator	9a82325	fix(ledger): split a full NFToken page before inserting
15:26	xrpl-validator	549abb1	fix(probe): hydrate trust lines for the currencies named in Paths
16:16	xrpl-validator	d5a08c8	fix(ledger): align IOU addition the way STAmount does
23:08	xrpl-validator	4bac28e	fix(ledger): apply tfSetfAuth to the trust line
23:52	xrpl-validator	be244f0	fix(ledger): CredentialCreate joins both owner directories